

Calculus 2

2 Lecture 2

Series with non-negative terms

Definition 2.1 (from Calculus 1). A sequence $\{a_n\}_{n=1}^{\infty}$ is called

- *increasing* if $a_{n+1} > a_n$ for all $n \in \mathbb{N}$;
- *nondecreasing* if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$;
- *decreasing* if $a_{n+1} < a_n$ for all $n \in \mathbb{N}$;
- *nonincreasing* if $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$.

Any of the above sequences is called *monotonic*. Increasing and decreasing sequences are called *strictly monotonic*.

Claim 2.1. If, for all $n \in \mathbb{N}$, $a_n > 0$ (respectively, $a_n \geq 0$, $a_n < 0$, $a_n \leq 0$), then the sequence

$$\left\{ A_n = \sum_{k=1}^n a_k \right\}_{n=1}^{\infty}$$

is increasing (respectively, nondecreasing, decreasing, nonincreasing).

Proof. Obvious. □

Claim 2.2. Let $a_n \geq 0$ for every $n \in \mathbb{N}$. Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ is convergent} \right) \iff \left(\left\{ A_n = \sum_{k=1}^n a_k \right\}_{n=1}^{\infty} \text{ is bounded} \right).$$

By definition, a sequence $\{b_n\}_{n=1}^{\infty}$ is bounded if

$$\exists M > 0 \quad \forall n \in \mathbb{N} : |b_n| < M.$$

Proof. From left to right,

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \stackrel{\text{def}}{\implies} \left(\exists \lim_{n \rightarrow +\infty} A_n \right) \implies (\{A_n\}_{n=1}^{\infty} \text{ is bounded}),$$

where the last implication is a result from Calculus 1.

Conversely, suppose $\{A_n\}_{n=1}^{\infty}$ is bounded. Since $a_k \geq 0$, the sequence $\{A_n\}_{n=1}^{\infty}$ is nondecreasing. Hence, by the corresponding result from Calculus 1, the limit $\lim_{n \rightarrow +\infty} A_n$ exists. By definition, this means that $\sum_{k=1}^{\infty} a_k$ converges. □

Claim 2.3 (Cauchy Condensation Test). Let $\{a_k\}_{k=1}^{\infty}$ be a sequence such that

- $a_k \geq 0$ for every $k \in \mathbb{N}$;
- $a_{k+1} \leq a_k$ for every $k \in \mathbb{N}$.

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges} \right).$$

The series

$$a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots + 2^k a_{2^k} + \cdots$$

is called the *condensed series*.

Proof. Since $a_{k+1} \leq a_k$, we have

$$\begin{aligned} a_2 &\leq a_2 \leq a_1, \\ 2a_4 &\leq a_3 + a_4 \leq 2a_2, \\ 4a_8 &\leq a_5 + a_6 + a_7 + a_8 \leq 4a_4, \end{aligned}$$

and, in general,

$$2^n a_{2^{n+1}} \leq a_{2^{n+1}} + \cdots + a_{2^{n+1}} \leq 2^n a_{2^n}.$$

By adding these inequalities, we obtain

$$a_2 + 2a_4 + 4a_8 + \cdots + 2^n a_{2^{n+1}} \leq \sum_{k=2}^{2^{n+1}} a_k \leq a_1 + 2a_2 + 4a_4 + \cdots + 2^n a_{2^n}. \quad (2.1)$$

Let

$$S_n = a_1 + 2a_2 + 4a_4 + \cdots + 2^n a_{2^n}$$

be the n -th partial sum of the condensed series $\sum_{k=0}^{\infty} 2^k a_{2^k}$, and let

$$A_n = a_1 + \cdots + a_n$$

be the n -th partial sum of $\sum_{k=1}^{\infty} a_k$. Then (2.1) can be rewritten as

$$\frac{1}{2}(S_{n+1} - a_1) \leq A_{2^{n+1}} - a_1 \leq S_n. \quad (2.2)$$

Since the sequences $\{A_n\}_{n=1}^{\infty}$ and $\{S_n\}_{n=1}^{\infty}$ are nondecreasing, inequality (2.2) implies

$$(\{A_n\}_{n=1}^{\infty} \text{ is bounded}) \iff (\{S_n\}_{n=1}^{\infty} \text{ is bounded}).$$

Applying Claim 2.2 to both series proves the result. $\square$

Claim 2.4 (The First Comparison Test). Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

$$0 \leq a_k \leq b_k \quad \text{for all } k \in \mathbb{N}.$$

Then the following implications hold:

1)

$$\left(\sum_{k=1}^{\infty} b_k \text{ converges} \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right) \implies \left(\sum_{k=1}^{\infty} b_k \text{ diverges} \right).$$

Proof. 1) Let

$$A_n = \sum_{k=1}^n a_k, \quad B_n = \sum_{k=1}^n b_k$$

be the n -th partial sums. Since $0 \leq a_k \leq b_k$ for every $k \in \mathbb{N}$, it is clear that

$$0 \leq A_n \leq B_n \quad (\forall n \in \mathbb{N}).$$

If $\sum_{k=1}^{\infty} b_k$ converges, then by Claim 2.2 the sequence $\{B_n\}_{n=1}^{\infty}$ is bounded. Hence $\{A_n\}_{n=1}^{\infty}$ is bounded, and Claim 2.2 gives the convergence of $\sum_{k=1}^{\infty} a_k$.

2) Assume that the statement is false. Then there are series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ such that

$$\sum_{k=1}^{\infty} a_k \text{ diverges} \quad \text{and} \quad \sum_{k=1}^{\infty} b_k \text{ converges}.$$

But, due to item 1), convergence of $\sum b_k$ implies convergence of $\sum a_k$. Thus we arrive at a contradiction. $\square$

Claim 2.5 (The Second Comparison Test). Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

- a) $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$;
- b) there exist $m > 0$ and $M > 0$ such that, for every $n \in \mathbb{N}$,

$$m \leq \frac{a_n}{b_n} \leq M.$$

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=1}^{\infty} b_k \text{ converges} \right).$$

Proof. Since $b_k > 0$, for every $k \in \mathbb{N}$ we have

$$\left(m \leq \frac{a_k}{b_k} \leq M \right) \implies (mb_k \leq a_k \leq Mb_k).$$

Therefore,

$$\sum_{k=1}^{\infty} a_k \text{ converges} \implies \sum_{k=1}^{\infty} mb_k \text{ converges} \implies \sum_{k=1}^{\infty} b_k \text{ converges},$$

where the first implication follows from [item 1\)](#) of [Claim 2.4](#) and the second from $m > 0$. Similarly,

$$\sum_{k=1}^{\infty} b_k \text{ converges} \implies \sum_{k=1}^{\infty} Mb_k \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges},$$

using [item 1\)](#) of [Claim 2.4](#) and $Mb_k \geq a_k$. □

Claim 2.6. Let $\{a_k\}_{k=1}^{\infty}$ and $\{b_k\}_{k=1}^{\infty}$ be two sequences such that

- a) $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$;
- b)

$$\lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lambda \in (0, +\infty).$$

Then

$$\left(\sum_{k=1}^{\infty} a_k \text{ converges} \right) \iff \left(\sum_{k=1}^{\infty} b_k \text{ converges} \right).$$

Proof. The condition

$$\lim_{k \rightarrow +\infty} \frac{a_k}{b_k} = \lambda \in (0, +\infty)$$

implies that

$$m \leq \frac{a_k}{b_k} \leq M$$

for some $m, M > 0$ (see Calculus 1). Therefore the statement follows from [Claim 2.5](#). □

Claim 2.7 (p -series convergence). The series

$$\sum_{k=1}^{\infty} \frac{1}{k^p}, \quad p \in \mathbb{R},$$

is called the p -series and converges if and only if $p > 1$.

Proof. If $p \leq 0$, then

$$\lim_{k \rightarrow +\infty} \frac{1}{k^p} \neq 0.$$

Therefore, due to the necessary condition for convergence ([Claim 1.5](#)), the series diverges.

If $p > 0$, then

$$\frac{1}{(k+1)^p} < \frac{1}{k^p} \quad (\forall k \in \mathbb{N}),$$

and all conditions of the Cauchy condensation test ([Claim 2.3](#)) are satisfied. Thus

$$\begin{aligned} \left(\sum_{k=1}^{\infty} \frac{1}{k^p} \text{ converges} \right) &\iff \left(\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} \text{ converges} \right) \\ &\iff \left(\sum_{k=0}^{\infty} \left(2^{1-p} \right)^k \text{ converges} \right) \\ &\iff 2^{1-p} < 1 \\ &\iff p > 1. \end{aligned}$$

□

Claim 2.8 (The Integral Test). Let

$$f : [1, +\infty) \rightarrow \mathbb{R}$$

be a positive, nonincreasing, continuous function. Then

$$\left(\sum_{k=1}^{\infty} f(k) \text{ converges} \right) \iff \left(\int_1^{+\infty} f(x) dx \text{ converges} \right).$$

Note. In general,

$$\sum_{k=1}^{\infty} f(k) \neq \int_1^{+\infty} f(x) dx.$$

Proof. For every $k \in \mathbb{N}$ and every $x \in [k, k+1]$, due to the monotonicity of f ,

$$f(k+1) \leq f(x) \leq f(k).$$

The geometric meaning of these inequalities is illustrated below.

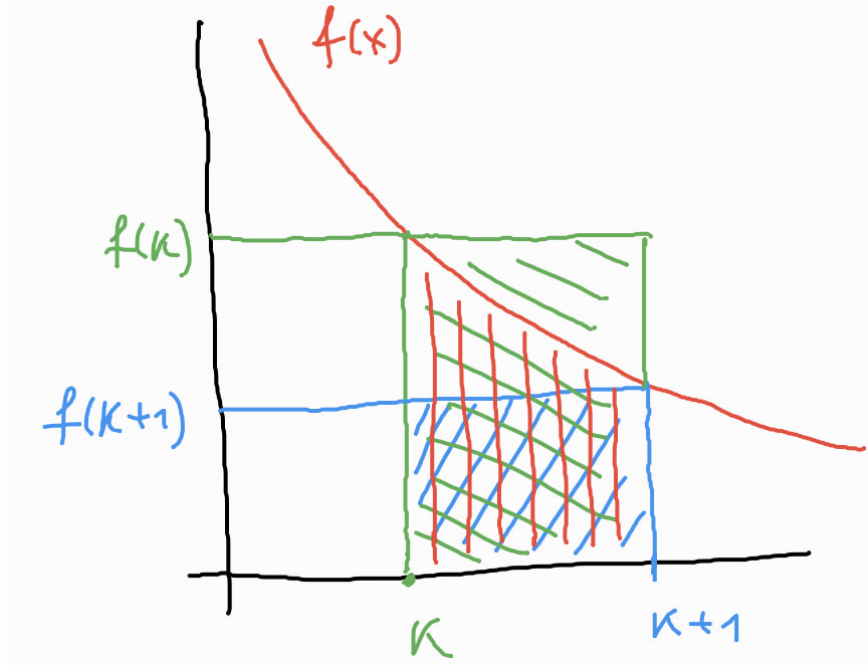


Figure 2.1

By integrating the inequalities over $[k, k+1]$, we obtain

$$f(k+1) = \int_k^{k+1} f(k+1) dx \leq \int_k^{k+1} f(x) dx \leq \int_k^{k+1} f(k) dx = f(k).$$

The integrals exist because f is continuous. By adding these inequalities from $k = 1$ to $k = n-1$, and writing

$$S_n = \sum_{k=1}^n f(k),$$

we obtain

$$S_n - f(1) = \sum_{k=1}^{n-1} f(k+1) \leq \sum_{k=1}^{n-1} \int_k^{k+1} f(x) dx = \int_1^n f(x) dx \leq \sum_{k=1}^{n-1} f(k) = S_{n-1}.$$

Hence

$$(\{S_n\}_{n=1}^{\infty} \text{ is bounded}) \iff \left(\left\{ \int_1^n f(x) dx \right\}_{n=1}^{\infty} \text{ is bounded} \right).$$

By [Claim 2.2](#), the left-hand condition is equivalent to the convergence of $\sum_{k=1}^{\infty} f(k)$. Since the sequence $\{\int_1^n f(x) dx\}$ is nondecreasing, the right-hand condition is equivalent to the convergence of $\int_1^{+\infty} f(x) dx$ (see Calculus 1). $\square$

Claim 2.9 (The Root Test). Let $a_k \geq 0$ for every $k \in \mathbb{N}$. Then:

1)

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} < 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} > 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Definition (see Calculus 1). Let $\{b_k\}_{k=1}^{\infty} \subset \mathbb{R}$ be a sequence. Then

$$\overline{\lim}_{n \rightarrow +\infty} b_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \left(\sup_{k \geq n} b_k \right)$$

is called the *upper limit* of $\{b_k\}$, and

$$\underline{\lim}_{n \rightarrow +\infty} b_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \left(\inf_{k \geq n} b_k \right)$$

is called the *lower limit* of $\{b_k\}$.

Proof. 1) Due to the definition of the upper limit,

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} < 1 \right) \implies \left(\exists q \in (0, 1) \exists N \in \mathbb{N} : \forall n \geq N, \quad 0 \leq a_n^{1/n} < q \right).$$

Equivalently, $0 \leq a_n < q^n$ for $n \geq N$. Since $q \in (0, 1)$, the geometric series $\sum_{k=0}^{\infty} q^k$ converges. By changing a finite number of terms (Claim 1.1), its tail $\sum_{k=N}^{\infty} q^k$ converges. Due to Claim 2.4 and $0 \leq a_n \leq q^n$ for $n \geq N$, the series $\sum_{k=N}^{\infty} a_k$ converges, and again by Claim 1.1, so does $\sum_{k=1}^{\infty} a_k$.

2) Due to the definition of the upper limit,

$$\left(\overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} > 1 \right) \implies \left(\forall N \in \mathbb{N} \exists n \geq N : a_n > 1 \right).$$

Thus $\lim_{k \rightarrow +\infty} a_k \neq 0$, and Claim 1.5 implies that $\sum_{k=1}^{\infty} a_k$ diverges. □

Claim 2.10. If $a_k > 0$ for every $k \in \mathbb{N}$, then

$$\underline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} \leq \underline{\lim}_{k \rightarrow +\infty} a_k^{1/k} \leq \overline{\lim}_{k \rightarrow +\infty} a_k^{1/k} \leq \overline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k}.$$

Proof. Without proof. □

Claim 2.11 (The Ratio Test). Let $a_k > 0$ for every $k \in \mathbb{N}$. Then:

1)

$$\left(\overline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} < 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\underline{\lim}_{k \rightarrow +\infty} \frac{a_{k+1}}{a_k} > 1 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. Follows directly from Claim 2.9 and Claim 2.10. □

Claim 2.12. The root test (Claim 2.9) is stronger than the ratio test (Claim 2.11).

Proof. Due to Claim 2.10, the root test is not weaker than the ratio test. Consider the series

$$\sum_{k=1}^{+\infty} 2^{(-1)^k - k}.$$

It ought to be clear that

$$\overline{\lim}_{k \rightarrow +\infty} \left(2^{(-1)^k - k} \right)^{1/k} = 2^{-1} < 1,$$

so by [Claim 2.9](#) the series converges. On the other hand,

$$\overline{\lim}_{k \rightarrow +\infty} \frac{2^{(-1)^{k+1}-k-1}}{2^{(-1)^k-k}} = \overline{\lim}_{k \rightarrow +\infty} 2^{(-1)^{k+1}-(-1)^k-1} = 2 > 1,$$

whereas

$$\underline{\lim}_{k \rightarrow +\infty} \frac{2^{(-1)^{k+1}-k-1}}{2^{(-1)^k-k}} = \underline{\lim}_{k \rightarrow +\infty} 2^{(-1)^{k+1}-(-1)^k-1} = 2^{-3} < 1.$$

Thus [Claim 2.11](#) cannot be applied. Since the root test works for this series and the ratio test does not, the claim is proved. $\square$

Claim 2.13 (Kummer's Test). Let $a_k > 0$ and $b_k > 0$ for every $k \in \mathbb{N}$. Let

$$c_k = \frac{a_k}{a_{k+1}} b_k - b_{k+1} \quad \text{for every } k \in \mathbb{N}.$$

Then:

1)

$$(\exists L > 0 \exists N \in \mathbb{N} : \forall n \geq N, c_n \geq L) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$\left(\sum_{k=1}^{\infty} \frac{1}{b_k} \text{ diverges and } \exists N \in \mathbb{N} : \forall n \geq N, c_n \leq 0 \right) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. 1) Without loss of generality, we can assume that

$$c_k \geq L \quad \forall k \in \mathbb{N}.$$

Since $a_k > 0$, we have

$$\left(\frac{a_k}{a_{k+1}} b_k - b_{k+1} \geq L \right) \iff (a_k b_k - a_{k+1} b_{k+1} \geq L a_{k+1}). \quad (*)$$

For every $n \in \mathbb{N}$, let

$$S_n = \sum_{k=1}^n a_k.$$

Then

$$\begin{aligned} S_n &= \frac{1}{L} \sum_{k=1}^n (L a_k) \\ &\leq \frac{1}{L} \sum_{k=1}^n (a_k b_k - a_{k+1} b_{k+1}) \\ &= \frac{1}{L} (a_1 b_1 - a_{n+1} b_{n+1}) \\ &\leq \frac{1}{L} a_1 b_1. \end{aligned}$$

Thus $\{S_n\}_{n=1}^{\infty}$ is bounded. Since $\{S_n\}$ is increasing, this implies that $\sum_{k=1}^{\infty} a_k$ converges.

2) Without loss of generality, we can assume that

$$c_k = \frac{a_k}{a_{k+1}} b_k - b_{k+1} \leq 0 \quad \forall k \in \mathbb{N}.$$

For every $k \in \mathbb{N}$,

$$\left(\frac{a_k}{a_{k+1}} b_k - b_{k+1} \leq 0 \right) \iff \left(\frac{a_k}{a_{k+1}} \leq \frac{b_{k+1}}{b_k} \right), \quad (**)$$

where $b_k > 0$. Multiplying the inequalities [\(**\)](#) from $k = 1$ to $k = n$, we obtain

$$\frac{a_1}{a_{n+1}} = \frac{a_1}{a_2} \frac{a_2}{a_3} \dots \frac{a_n}{a_{n+1}} \leq \frac{b_2}{b_1} \frac{b_3}{b_2} \dots \frac{b_{n+1}}{b_n} = \frac{b_{n+1}}{b_1}.$$

Thus

$$\frac{a_1}{a_{n+1}} \leq \frac{b_{n+1}}{b_1} \iff a_{n+1} \geq a_1 b_1 \frac{1}{b_{n+1}}.$$

Since

$$\sum_{k=1}^{\infty} a_1 b_1 \frac{1}{b_k}$$

diverges (it is given that $\sum 1/b_k$ diverges and $a_1 b_1$ is a nonzero constant), [item 2\)](#) of [Claim 2.4](#) implies that $\sum_{k=1}^{\infty} a_k$ diverges. $\square$

Claim 2.14 (Bertrand's Test). Let $a_k > 0$ for every $k \in \mathbb{N}$. Let

$$\lim_{k \rightarrow +\infty} \ln(k) \left(k \left(\frac{a_k}{a_{k+1}} - 1 \right) - 1 \right) = B \in [-\infty, +\infty].$$

Then:

1)

$$(B > 1) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$(B < 1) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

The cases $B = \pm\infty$ are allowed in this claim.

Proof. **1)** Let

$$b_k = k \ln k \quad (k \geq 2).$$

Then

$$\begin{aligned} c_k &= \frac{a_k}{a_{k+1}} b_k - b_{k+1} \\ &= \ln(k) \left(k \left(\frac{a_k}{a_{k+1}} - 1 \right) - 1 \right) - (k+1)(\ln(k+1) - \ln k). \end{aligned} \quad (***)$$

The first term tends to B , and

$$\lim_{k \rightarrow +\infty} (k+1)(\ln(k+1) - \ln k) = 1$$

(see Calculus 1). Hence, if $B > 1$,

$$\lim_{k \rightarrow +\infty} c_k = B - 1 > 0.$$

Therefore

$$\exists L > 0 \exists N \in \mathbb{N} : \forall n \geq N, c_n \geq L,$$

and [item 1\)](#) of [Claim 2.13](#) implies that $\sum_{k=1}^{\infty} a_k$ converges.

2) Since

$$\int_2^{+\infty} \frac{1}{x \ln x} dx = \ln \ln x \Big|_2^{+\infty}$$

diverges, [Claim 2.8](#) gives that the series

$$\sum_{k=2}^{+\infty} \frac{1}{k \ln k}$$

diverges. If $B < 1$, then by equality [\(***\)](#),

$$\lim_{k \rightarrow +\infty} c_k = B - 1 < 0,$$

so there exists N such that $c_n < 0$ for every $n \geq N$. Therefore, due to [item 2\)](#) of [Claim 2.13](#), the series $\sum_{k=1}^{\infty} a_k$ diverges. $\square$

Claim 2.15 (The Gauss Test). Let

- a) $a_k > 0$ for every $k \in \mathbb{N}$;
- b) there exist $\varepsilon > 0$ and $\lambda, \mu \in \mathbb{R}$ such that, for every $k \in \mathbb{N}$,

$$\frac{a_k}{a_{k+1}} = \lambda + \frac{\mu}{k} + \frac{\gamma_k}{k^{1+\varepsilon}},$$

where the sequence $\{\gamma_n\}_{n=1}^{\infty}$ is bounded.

Then:

1)

$$((\lambda > 1) \text{ or } (\lambda = 1 \text{ and } \mu > 1)) \implies \left(\sum_{k=1}^{\infty} a_k \text{ converges} \right);$$

2)

$$((\lambda < 1) \text{ or } (\lambda = 1 \text{ and } \mu \leq 1)) \implies \left(\sum_{k=1}^{\infty} a_k \text{ diverges} \right).$$

Proof. Follows directly from [Claim 2.14](#) (can you see why?). $\square$

Claim 2.16. Bertrand's test ([Claim 2.14](#)) is stronger than the Gauss test ([Claim 2.15](#)).

Proof. Since [Claim 2.15](#) follows from [Claim 2.14](#), Bertrand's test is not weaker than the Gauss test. One can verify by direct calculation that [Claim 2.14](#) works for

$$\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$$

and [Claim 2.15](#) does not. $\square$